Daily Quiz 13 (ELL782/ELL7282) - Computer Architectures

Instructions

- **Time**: 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.

Points: 10/10

Name *

test

Entry Number *

test

✓ **Correct** 1/1 Points

Consider [1] add r1, r2, r3 followed immediately by [2] sub r4, r1, r2 in the 5-stage pipeline, with [1] in IF in cycle 1 and no bubbles. When does [2] actually need r1, and when does [1] actually produce it?

- Needed in cycle 4 (EX); produced in cycle 5 (RW)
- Needed in cycle 3 (OF); produced in cycle 5 (RW)

- (●) Needed in cycle 4 (EX); produced at the end of cycle 3 (EX)
- ( ) Needed in cycle 3 (OF); produced at the end of cycle 3 (EX)

✓ **Correct** 1/1 Points

Which statements correctly describe the rules and choices for forwarding paths in this pipeline?

Select all that apply.

- [x] A forwarding path always goes from a later pipeline stage to an earlier stage.
- [x] The 2-stage path MA → OF is not required.
- [ ] The IF stage receives forwarded values so that branch targets can be computed early.
- [x] Paths are added as late as possible; for example, EX → OF is avoided because MA → EX exists.

✓ **Correct** 1/1 Points

In the MA stage with forwarding, the EX-MA latch supplies aluResult and op2, and a multiplexer M6 sits in front of the memory unit with an additional input coming from RW. What does M6 select?

- (●) The store data: either op2 from the EX-MA latch or the value forwarded from RW.
- ( ) The memory address: either aluResult or the value forwarded from RW.
- ( ) The next PC: either PC + 4 or the value forwarded from RW.
- ( ) The register-write value: either the data memory output or aluResult.

✓ **Correct** 1/1 Points

Consider the sequence [1] ld r1, 4[r2], [2] st r4, 10[r3], [3] st r5, 10[r6], [4] sub r7, r1, r2, issued one per cycle with [1] in IF in cycle 1. In which cycle, and through which path, does [4] receive r1?

- Cycle 6, through RW → EX
- No forwarding is needed; [4] reads the updated r1 from the register file in cycle 5.
- Cycle 4, through MA → EX
- Cycle 5, through RW → OF

✓ **Correct** 1/1 Points

With all forwarding paths present, consider [1] ld r1, 10[r2] followed immediately by [2] sub r4, r1, r2. What must the pipeline do?

- Stall [2] for three cycles until [1] finishes RW, then read r1 from the register file.
- Hold [2] in OF for one cycle (a bubble goes into EX), then forward r1 through RW → EX.
- Forward r1 through EX → OF in cycle 3 without any stall.
- Forward r1 through MA → EX in cycle 4 without any stall.

✓ **Correct** 1/1 Points

How do the interlocks change once full forwarding is added to the pipeline?

- Data-lock checks only the load-use case (1-cycle stall); branch-lock stays the same as before.
- Data-lock checks only the load-use case; branch-lock now invalidates only the IF stage.
- Both data-lock and branch-lock are no longer needed.
- Data-lock still checks every RAW conflict between OF and EX/MA/RW; branch-lock is removed.

✓ **Correct** 1/1 Points

Instruction A = add r5, r1, r2 is in EX. The MA stage holds sub r1, r3, r4 and the RW stage holds add r1, r6, r7. Which value of r1 must the forwarding unit supply to A?

- ○ The value from RW (add r1, r6, r7), because that instruction has finished computing.
- ○ The value read from the register file in OF; forwarding is not needed.
- ◉ The value from MA (sub r1, r3, r4), the latest instruction that is earlier than A.
- ○ Neither; A must stall because two producers conflict on r1.

✓ **Correct** 1/1 Points

In the pipeline with forwarding, which (EX-stage instruction, OF-stage instruction) pairs make the data-lock unit stall the pipeline?

Select all that apply.

- ☐ EX: ld r3, 0[r5]; OF: mov r3, 8
- ☐ EX: add r3, r4, r5; OF: sub r6, r3, r7
- ☑ EX: ld r3, 0[r5]; OF: ld r6, 4[r3]
- ☑ EX: ld r3, 0[r5]; OF: add r6, r7, r3

✓ **Correct** 1/1 Points

A compiler applies dead-code removal and function inlining, cutting a program's dynamic instruction count by 20%. On the same processor (same f), the CPI goes from 1.0 to 1.1. What is the performance of the optimized program compared with the original?

- ○ 1.25× faster
- ○ The same performance
- ◉ About 1.14× faster

About 0.88× (slower)

✓ **Correct** 1/1 Points

A program binary has a loop body of 5 static instructions (including the loop branch) that runs 100 times, plus 2 static setup instructions that run once. What value of #insts should be used in the performance equation?

700

7

500

502

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